

Assignment - 2

Part - I

$$1. (a) f(n_1, n_2) = e^{-\lambda} \frac{\lambda^{n_1}}{n_1!} \binom{n_1}{n_2} p^{n_2} (1-p)^{n_1-n_2} \quad 0 \leq n_2 \leq n_1$$

$$f_{X_1}(n_1) = \sum_{n_2=0}^{n_1} f(n_1, n_2)$$

$$f_{X_1}(n_1) = \sum_{n_2=0}^{n_1} e^{-\lambda} \frac{\lambda^{n_1}}{n_1!} \binom{n_1}{n_2} p^{n_2} (1-p)^{n_1-n_2}$$

$$= e^{-\lambda} \frac{\lambda^{n_1}}{n_1!} \sum_{n_2=0}^{n_1} \binom{n_1}{n_2} p^{n_2} (1-p)^{n_1-n_2}$$

$$\text{binomial theorem: } \sum_{n_2=0}^{n_1} \binom{n_1}{n_2} p^{n_2} (1-p)^{n_1-n_2} = (1+p)^{n_2} = 1$$

$$\therefore f_{X_1}(n_1) = e^{-\lambda} \frac{\lambda^{n_1}}{n_1!}$$

$$X_1 \sim \text{Poisson}(\lambda)$$

$$f_{X_2}(n_2) = \sum_{n_1=n_2}^{\infty} f(n_1, n_2)$$

$$= \sum_{n_1=n_2}^{\infty} e^{-\lambda} \frac{\lambda^{n_1}}{n_1!} \frac{n_1!}{n_2! (n_1-n_2)!} p^{n_2} (1-p)^{n_1-n_2}$$

$$f_{X_2}(n_2) = e^{-\lambda} \frac{p^{n_2}}{n_2!} \sum_{n_1=n_2}^{\infty} \frac{\lambda^{n_1} (1-p)^{n_1-n_2}}{(n_1-n_2)!}$$

$$\text{let } K = n_1 - n_2$$

$$n_1 = K + n_2$$

$$f_{X_2}(n_2) = e^{-\lambda} \frac{(\lambda p)^{n_2}}{n_2!} \sum_{K=0}^{\infty} \frac{(\lambda(1-p))^K}{K!}$$

$$\sum_{K=0}^{\infty} \frac{a^K}{K!} = e^a$$

$$f_{X_2}(n_2) = e^{-\lambda} \frac{(\lambda p)^{n_2}}{n_2!} e^{\lambda(1-p)}$$

$$= e^{-\lambda p} \frac{(\lambda p)^{n_2}}{n_2!}$$

$$X_2 \sim \text{Poisson}(\lambda p)$$

$$(b) \quad Y = X_1 - X_2$$

$$X_1 = X_2 + Y$$

$$\text{joint PMF of } (X_2, Y) : f_{X_2, Y}(n_2, y) = f(n_2 + y, n_2)$$

$$f_{X_2, Y}(n_2, y) = e^{-\lambda} \frac{\lambda^{n_2+y}}{(n_2+y)!} \binom{n_2+y}{n_2} p^{n_2} (1-p)^y$$

$$= e^{-\lambda} \frac{\lambda^{n_2+y}}{n_2! y!} p^{n_2} (1-p)^y$$

$$= e^{-\lambda p} \frac{(\lambda p)^{n_2}}{n_2!} e^{-\lambda(1-p)} \frac{(\lambda(1-p))^y}{y!}$$

$$f_{X_2, Y}(n_2, y) = f_{X_2}(n_2) f_Y(y)$$

$$X_2 \sim \text{Poisson}(\lambda p)$$

$$Y \sim \text{Poisson}(\lambda(1-p))$$

two discrete random variables A, B are independent if

$$f_{A,B}(a,b) = f_A(a) f_B(b)$$

Thus $\therefore X_2$ and Y independent Poisson RV

$$2. \quad P(\text{false coin}) = \frac{m}{n}$$

$$P(\text{true coin}) = 1 - \frac{m}{n}$$

for each coin i , $x_i = \begin{cases} 1 & \text{selected coin is false} \\ 0 & \text{selected coin is true} \end{cases}$

$$P(x_i = 1) = \frac{m}{n}$$

$$x_i \sim \text{Bernoulli}\left(\frac{m}{n}\right)$$

$$P(x_i = 0) = 1 - \frac{m}{n}$$

let x denote total number of false coin

$$X = \sum_{i=1}^n x_i$$

$$X \sim \text{Binomial}\left(n, \frac{m}{n}\right)$$

$$P(X=r) = {}^nC_r \left(\frac{m}{n}\right)^r \left(1 - \frac{m}{n}\right)^{n-r}$$

$$\text{as } n \rightarrow \infty$$

$$p = \frac{m}{n} \rightarrow 0$$

$$np = m$$

$$\text{binomial } (n, \frac{m}{n}) \longrightarrow \text{Poisson } (m)$$

$$P(X=r) = \frac{e^{-m} m^r}{r!}$$

$$3. (a) \int_0^\infty \int_0^{n_1} c e^{-n_1} dn_2 dn_1 = 1$$

$$\int_0^\infty c n_1 e^{-n_1} dn_1 = 1$$

$$\int_0^\infty n_1 e^{-n_1} dn_1 = \left[-n_1 e^{-n_1} \right]_0^\infty - \int_0^\infty -e^{-n_1} dn_1 = \left[-e^{-n_1} \right]_0^\infty = 1$$

$$\therefore c = 1$$

$$(b) f_{X_2/X_1}(n_2/n_1) = \frac{f(n_1, n_2)}{f_{X_1}(n_1)}$$

$$f_{X_1}(n_1) = \int_0^{n_1} e^{-n_1} dn_2 = n_1 e^{-n_1} \quad n_1 \geq 0$$

$$f_{X_2/X_1}(n_2/n_1) = \frac{e^{-n_1}}{n_1 e^{-n_1}} = \begin{cases} \frac{1}{n_1} & 0 \leq n_2 \leq n_1 \\ 0 & \text{otherwise} \end{cases}$$

(c) need to find $P(X_2 > 1 | X_1 = 2)$

$$f_{X_2/X_1}(x_2/2) = \begin{cases} \frac{1}{2} & 0 \leq x_2 \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$P(X_2 > 1 | X_1 = 2) = \int_1^2 \frac{1}{2} dx_2 = \frac{1}{2}$$

4. $P(N=n) = \frac{e^{-\lambda} \lambda^n}{n!}$

$$E[X_i] = \mu \quad \text{Var}(X_i) = \sigma^2$$

$$S = \sum_{i=1}^N X_i \quad \text{with } S=0 \text{ when } N=0$$

law of total Expectation:

$$E(S) = E(E(S|N))$$

$$E(S|N=n) = E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n) = n\mu$$

$$E(S|N) = N\mu$$

Now,

$$E(S) = E(E(S|N)) = E(N\mu) = \mu E(N) = \underline{\underline{\mu\lambda}}$$

law of total variance:

$$\text{Var}(S) = E(\text{Var}(S|N)) + \text{Var}(E(S|N))$$

$$\text{Var}(S|N=n) = n\sigma^2$$

$$\text{Var}(S|N) = N\sigma^2$$

$$E(\text{Var}(S|N)) = \sigma^2 E(N) = \sigma^2\lambda$$

$$\text{Var}(E(S|N)) = \text{Var}(N\mu) = \mu^2 \text{Var}(N) = \mu^2\lambda$$

$$\text{Var}(S) = \sigma^2\lambda + \mu^2\lambda = (\sigma^2 + \mu^2)\lambda$$

5. Covariance formula: $\text{Cov}(X, Y) = E((X - E(X))(Y - E(Y)))$
 $= E(XY) - E(X)E(Y)$

X and Y are uncorrelated if $\text{Cov}(X, Y) = 0$

(a) $\text{Cov}(U, V) = \text{Cov}(X_1 + X_2, X_1 - X_2)$
 $= \text{Cov}(X_1^2) - \text{Cov}(X_2^2)$
 $= \text{Var}(X_1^2) - \text{Var}(X_2^2)$
 $= 0$

U and V are uncorrelated

(b) let X and $Y = X^2$ $X \in \{-1, 0, 1\}$
 $\text{Cov}(X, X^2) = E(X^3) - E(X)E(X^2)$
 $= 0 - 0$
 $= 0$

so X and X^2 are uncorrelated

but X^2 is dependent on X

6. let both Urn 1 and Urn 2 contain T balls
 w_1 white balls in Urn 1
 w_2 white balls in Urn 2

$$p = P(\text{white from Urn 1}) = \frac{w_1}{T}$$

$$q = P(\text{white from Urn 2}) = \frac{w_2}{T}$$

$$p^n = q^n + (1-q)^n \quad \text{and} \quad 0 \leq p, q \leq 1$$

$$0 \leq q \leq 1$$

$$0 \leq 1-q \leq 1$$

$$\text{for } n \geq 1 \quad q^n \leq q$$

$$(1-q)^n \leq (1-q)$$

$$q^n + (1-q)^n \leq 1$$

therefore, for all $q \in [0, 1]$ there exists a p

$$p = (q^n + (1-q)^n)^{1/n} \leq 1$$